



MFCA: Multimodal Object Detection Based on Feature Calibration and Aggregation

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Abstract. Multimodal object detection has recently attracted much attention. By integrating feature from various modalities, the detection models' accuracy can be significantly enhanced. Despite this, current approaches predominantly emphasize the straightforward fusion of multimodal features, often neglecting the redundant information, such as noise, that can be present within individual modal features. This can lead to models that fail to effectively capture important features of the object, leading to inadequate detection results. To address this problem, we propose a multimodal object detection model based on feature calibration and aggregation, termed MFCA, which comprises a Spatial Calibration Module (SCM) and a Channel Aggregation Module (CAM). The aim is to focus on the information of a mono-modal and supplement it with the valid information from another modality. SCM filters out the redundant information from the other modality and incorporates the valid information into this modality, which preventing errors in features due to incorporation of redundant information. Then, CAM replaces the unimportant feature channels in one modality using another modality to enrich the detailed semantic information of the features. By jointly fusing multimodal features across different dimensions on the same scale, the ability of model to perceive efficient features is enhanced. Experiments performed on the KAIST, FLIR, and VEDAI datasets validate the effectiveness of the proposed scheme, demonstrating that it outperforms other state-of-the-art methods.

Keywords: Multimodal object detection · Spatial calibration · Channel aggregation · Feature fusion

1 Introduction

Object detection is an inextricable branch of computer vision and has been extensively employed in real-world applications. However, traditional object detection, constrained by environmental factors, is occasionally inadequate for all real-world scenarios. For example, in low-light conditions, the quality of RGB images may degrade substantially, which in turn impacts detection accuracy. In tasks

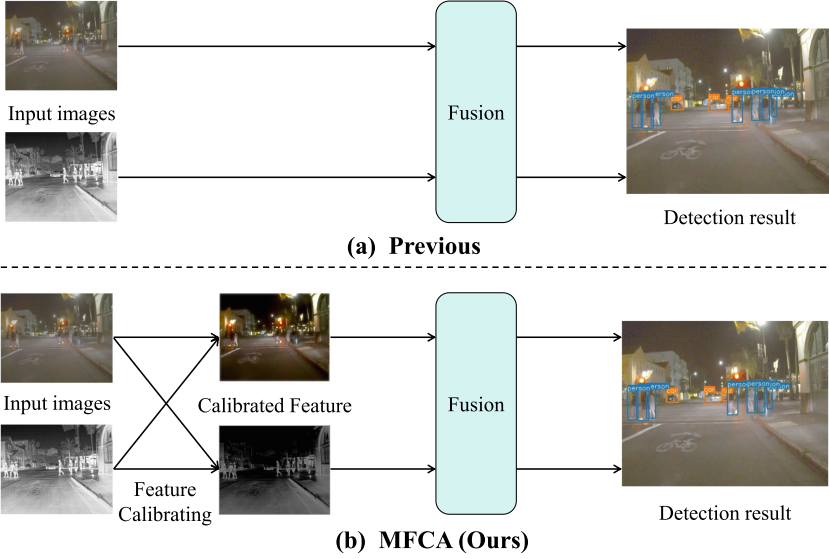


Fig. 1. (a) Previous: directly fuse features. (b) MFCA: fuse the features after calibrating the feature information.

demanding high accuracy and robustness, such errors can have a significant impact. Therefore, fusing information from multimodal images has become a common approach to enhance model performance, helping achieve better accuracy and robustness. Combining RGB and infrared (IR) images takes advantage of their complementary properties, providing a beneficial solution. IR images make the ability of model to discern object outlines, especially in challenging scenarios with poor illumination or occlusion. The two modalities are mutually reinforcing, thereby improving the efficiency of the object detection model.

Cross-modal object detection has received significant attention, highlighting the need for effective cross-modal feature fusion methods in these algorithms. Early cross-modal fusion techniques often relied on fundamental operations like cascading, simple summation, and element-wise multiplication. For instance, in GFD-SSD [30], two novel gated fusion unit variants were proposed to explore the combination of feature mappings from two intermediate SSD layers. Zhou et al. [31] employed differential amplifiers to compare the consistencies and discrepancies among different modalities, effectively utilizing both common and differential modal information. Additionally, Kim et al. [10] addressed model uncertainties by utilizing a loss function aimed at aligning the features of the two different modalities. The above method focuses too much on the direct fusion of two modalities when fusing features from different modalities, without taking into account the redundant information present in a each modality. This information persists in the fused features, leading the model to concentrate on

unnecessary aspects and diminishing its capacity to recognize critical features. Figure 1 shows the difference between previous work and ours.

Therefore, we propose a multimodal feature calibration and aggregation (MFCA) method to address the above problems. MFCA separately extracts multimodal features with a parallel structure comprising SCM and CAM, which are cascaded to effectively capture both global and local information. The SCM is specifically designed to obtain the features of other modalities while retaining the original effective information of this modality to the greatest extent possible. This method aims to reduce the effect of other redundant information such as noise and improve the fusion of different modalities. To facilitate the fusion of semantic information across different modalities, we adopt the idea of channel switching [1] and design the CAM. The fused features are then processed by the subsequent detection head for final detection. Experimental results show that the proposed algorithm significantly outperforms commonly used methods.

The main contributions of this paper are as follows:

- A spatial calibration module is proposed, which filters the redundant information from another modality and incorporates the valid information into the current modality. This allows the interference caused by the redundant information to be calibrated to the features.
- A channel aggregation module is proposed, which analyses the features of both modalities in terms of channel dimension. Unimportant feature information is replaced with features of the other modality in order to enrich the semantic information.
- We perform comprehensive experiments on three different public datasets, which demonstrate the generalizability of our proposed model.

2 Related Work

2.1 Object Detection

Object detection methods are typically categorized into approaches based on transformers and those utilizing CNNs. Additionally, CNN-based methods are further classified into single-stage and two-stage. Among these, the R-CNN [5] series is the classic example of the two-stage methods. Faster R-CNN [18] utilizes a region proposal network to generate candidate regions. It eliminates the need for selective search algorithms of RCNN and improves the speed of detection.

Unlike traditional two-stage methods, one-stage detection frameworks such as YOLO [7] can predict the location and class directly from the image without explicitly generating candidate regions. Additionally, in contrast to the consideration of multiple scales when processing the input image, YOLO performs all object detection at the same scale. This design makes YOLO simpler, but also limits its effectiveness in small object detection.

Distinct from traditional CNN-based object detectors, DETR [2] treats the object detection task as a set prediction task, concentrating on detecting a fixed number of objects. The model generates the prediction set simultaneously by

leveraging interactions between the global image context and the objects. It utilizes a loss function based on two-part graph matching to align the predicted bounding boxes with the ground truth. This enables DETR to create a unique set of prediction boxes, ensuring each object is associated with only one prediction box.

Most existing multimodal object detection models are designed on the basis of traditional object detection models.

2.2 Object Detection with Multimodal Data Fusion

Recently, multimodal data has been applied in various practical scenarios widely, such as autonomous driving, saliency object detection, and remote sensing classification, among others [3, 25].

Combining image features from different modalities with complementary features can enhance the performance of various tasks. To fully leverage the information across different modalities, Sharma et al. proposed YOLOrs [19], which designed a medium-term fusion architecture using a concentrated region and element cross-product approach. Subsequently Fang et al. designed a method [16] via attention-based cross-modal fusion, where an attention map is inferred in parallel from the common and differential features of two modal pictures, which is subsequently integrated with a feature map to enable adaptive enhancement or selection of features. In addition, Wang et al. designed RISNet [23] to improve the complementary features from two modalities. It employs a mutual information minimization module to enable the network to fully utilize the complementary multimodal information, thereby improving object detection performance. Furthermore, to overcome the lack of existing methods in modeling remote dependencies between modalities, CFT [15] utilizes global attention mechanisms to enhance the quality of feature fusion. However, these methods simply fuse local and global information without fully considering the problem of redundant information present in mono-modal features.

Consequently, we propose the MFCA method which employs the mid-fusion strategy. This method considers the importance of mono-modal feature information, as well as spatial and channel dimension feature information across different modalities, to minimize interference from redundant information. Furthermore, to enrich the semantic information, we switch the channels that do not contain significant information from the other modality.

3 Methodology

The model we designed in this paper is adapted from YOLOv5 [7], the details of the model are shown in Fig. 2. It processes IR and RGB images using two CSPDarknet backbones. Each backbone integrates three MFCA modules for feature map fusion across different stages. An MFCA module comprises two distinct sub-modules: one for spatial calibration module and the other for channel aggregation module. In the spatial calibration module, we perform cross-modal

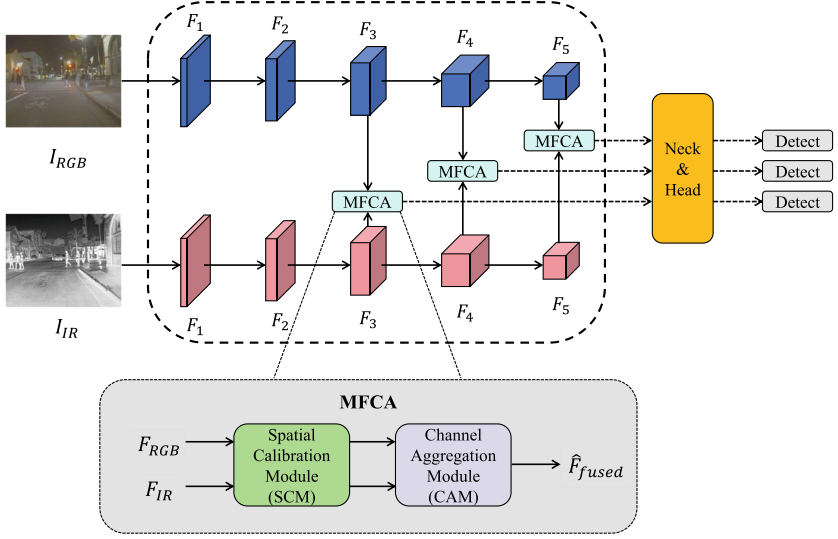


Fig. 2. The overview of our multimodal object detection framework. The RGB and IR feature extraction modules are represented by the top and bottom branches, respectively. $F_1 \sim F_5$ denote the feature maps at various scales. The MFCA module, shown at the bottom of the figure, demonstrates our proposed approach to feature fusion. The Neck & Head functions as a multiscale feature aggregation network, ultimately producing the final detection results.

global interaction using the attention mechanism, and then enable the interaction of spatial dimensional information through the MLP layer. This approach allows both modalities to focus efficiently on the complementary information they exchange, while mitigating the effects of uncertainty and noise from different modalities. After the spatial calibration, the channel aggregation module replaces channels with insignificant prediction information with corresponding channels from another modality. This process enables each modality to preserve its distinct features while efficiently integrating the semantic information from other modalities. After channel switching, hybrid channel embedding is used to generate enhanced output features. As a result, the feature maps are calibrated and fused in spatial and channel dimensions through multidimensional interactions, and the fused feature maps are sent to subsequent necks and detection heads to generate the final results.

3.1 Spatial Calibration Module

Due to the limited receptive field of convolutional operations, existing CNN-based multimodal fusion modules are difficult to adequately capture remote contextual information from the features of two modalities. Considering the powerful

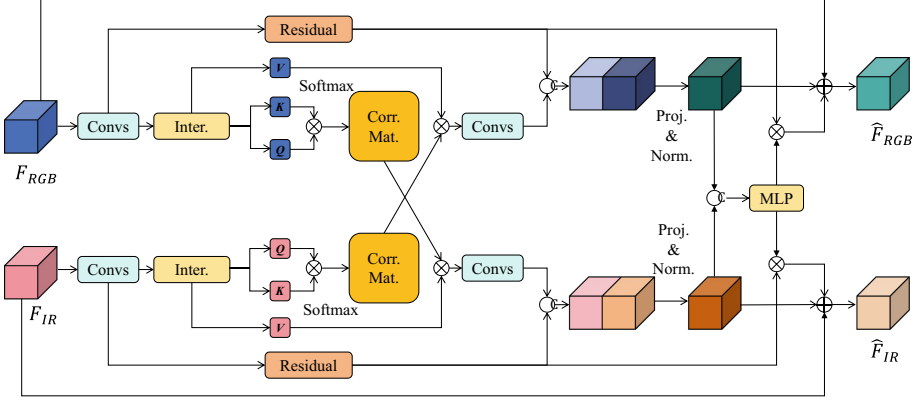


Fig. 3. Spatial calibration module (SCM): Fusion and calibration of feature information from both modalities along spatial dimensions. Inter. denotes interactive branch of feature. Corr. Mat. denotes the correlation matrix.

cross-modal global modeling capabilities of the attention mechanism, we propose an attention-based spatial calibration module to improve information interaction between the two modalities. The structure of the SCM is shown in Fig. 3.

The MFCA we proposed is a two-branch backbone network dedicated to feature extraction from RGB-IR image pairs. Firstly, feature extraction is performed separately for the RGB and IR images respectively, which is described in Eq. 1.

$$F_x = \text{Conv}_{3 \times 3}(\text{ReLU}(\text{Conv}_{1 \times 1}(X_{in}))) \quad (1)$$

where X_{in} denotes the input mono-modal RGB image or IR image, F_x denotes the feature information corresponding to the image. Then, the feature maps are chunked to generate Q, K, V for the corresponding modalities:

$$Q_x, K_x, V_x = \mathcal{F}_{\text{chunk}}(F_x) \quad (2)$$

The cross-attention mechanism employed to improve cross-modal feature fusion builds on traditional self-attention. However, while traditional self-attention encodes input vectors as Q, K, V and calculates global attention maps through matrix multiplication, this approach can result in high memory consumption. In contrast, Efficient Attention [21] uses a global context vector. Thereby, we design an efficient cross-modal attention mechanism as shown in Eq. 3:

$$\begin{cases} Z_1 = \text{Attention}(Q_1, K_1, V_2) = \text{softmax}\left(\frac{Q_1 K_1^T}{\sqrt{d_k}}\right) V_2 \\ Z_2 = \text{Attention}(Q_2, K_2, V_1) = \text{softmax}\left(\frac{Q_2 K_2^T}{\sqrt{d_k}}\right) V_1 \end{cases} \quad (3)$$

The correlation matrix is constructed using a dot-product operation to quantify the pairwise relationships among features. Subsequently, a softmax function is applied to normalize these correlation scores, representing the degree of similarity between distinct features across the RGB and IR modalities. Thereafter, the vectors Z_1 or Z_2 are derived by element-wise multiplication of the correlation matrix with a vector V , thereby refining the corresponding features and enhancing their representation through the exploitation of inter-modality similarities. Specifically, we use Q and K from the same modality, but V is from another modality. Our goal is to generate the attention map by evaluating the correlation of Q with K , and subsequently obtain the feature values from V using the derived attention graph. This cross-attention mechanism facilitates more effective information interaction between the two feature maps during the fusion task, thereby enhancing the overall fusion result.

While the aforementioned cross-modal attention mechanism effectively learns global weights for global calibration, we additionally incorporate spatial feature calibration to refine and correct local information. The *RGB* and *IR* inputs are spliced together in the spatial dimension after projection, and then processed through the MLP layer to generate two $H \times W \times 1$ attention tensors. Then, these tensors are merged with the original and interactive features to calibrate the two modalities together. The resulting combined features are subsequently forwarded to the next stage for fusion.

3.2 Channel Aggregation Module

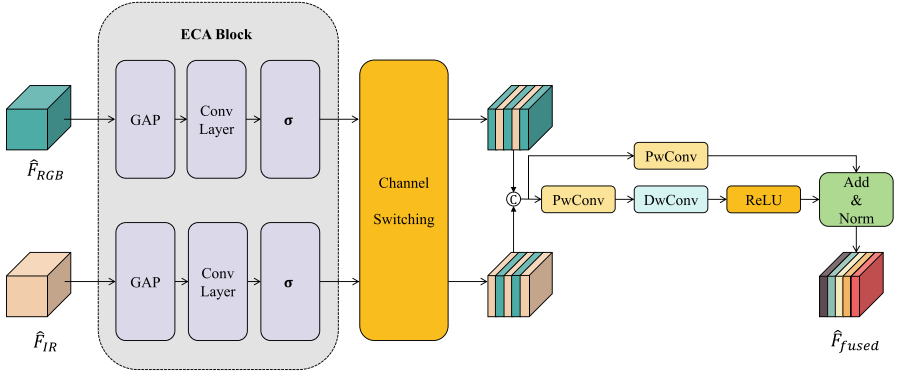


Fig. 4. Channel Aggregation module (CAM): Using channel switching and hybrid channel embedding to obtain detailed semantic information of different modal feature maps.

Considering the importance of focusing on channel-dimension features for obtaining rich semantic information, modules are allowed to learn shared features

between modalities while preserving unique mono-modal features. To enable attention to channel dimension features, we design a channel aggregation module that is efficient for multimodal interaction. The structure of the CAM is shown in Fig. 4.

To optimize the module’s efficiency, we utilize an ECA [22] block. This block comprises a global average pooling (GAP) layer, followed by a 1D convolution, and concludes with a sigmoid function. This design effectively captures local cross-channel interactions. The details are described as follows:

$$\omega_m = \sigma(f(\text{GAP}(X_m))), \quad (4)$$

where X_m represents the feature mapping from a particular modality with the dimensions of $H \times W \times C$. The term f refers to a convolutional layer, and σ denotes the sigmoid function. The ECA block transforms the input feature map into a $1 \times 1 \times C$ vector and employs a 1D convolution to facilitate cross-channel interactions. It then computes the weights ω_m using the sigmoid function. The output ω_m with dimensions $1 \times 1 \times C$, is subsequently used to perform channel-wise modulation. The process can be described as follows:

$$X_m = \begin{cases} X_m, & \text{if } \omega_m \geq k, \\ X_{m'}, & \text{if } \omega_m < k \end{cases}, \quad (5)$$

where m' denotes a feature from a different modality, and k represents a pre-defined threshold. If a channel’s weight falls below this threshold, the model replaces it with the corresponding channel from the alternative modality.

Subsequently, we employ a hybrid channel embedding to integrate features from both paths. To enhance robust detection during channel fusion, we leverage information from surrounding regions. Inspired by ConvMLP [13], we incorporate a pointwise convolution to create a skip-connected structure. This approach merges features of size $R^{H \times W \times 2C}$ into a final output of size $R^{H \times W \times C}$ for further processing in the network.

4 Experiments

In this section, we first present three datasets: KAIST, FLIR, and VEDAI. Following that, we outline the implementation details and analyze the experimental results, comparing the proposed MFCA method with current state-of-the-art techniques. Unless mentioned otherwise, all evaluations are performed using a standard setup. Lastly, we perform ablation studies specifically on the FLIR dataset.

4.1 Datasets

KAIST Dataset [6]. The KAIST dataset is a widely recognized benchmark designed for multispectral pedestrian detection, suitable for scenes with varying lighting conditions. It consists of 8,963 training image pairs and 2,252 testing

pairs, each with a resolution of 640×512 . Performance is typically measured using the log-average miss rate (MR) metric, which computes the average miss rate across nine false positives per image (FPPI) values, distributed evenly within the range $[10^{-2}, 1]$. A lower MR value signifies better detection accuracy.

FLIR Dataset [4]. The FLIR dataset serves as a stringent benchmark for multispectral object detection encompassing scenes from both daytime and nighttime. It includes 5,142 multispectral image pairs, with 4,129 for training and 1,013 for testing. The dataset comprises three object categories: “person,” “car,” and “bicycle.” For our experiments, we used the FLIR-aligned version [26]. Performance is evaluated using average precision (AP), a widely adopted metric in object detection. Classification accuracy and Intersection over Union (IoU), typically with a threshold of 0.5, are used to assess positive and negative samples. The mean Average Precision (mAP) is then calculated across all object classes, where higher AP and mAP values indicate stronger performance, in contrast to the MR^{-2} metric.

VEDAI Dataset [17]. The VEDAI dataset is a public resource for small object detection in remote sensing, with more than 3,700 annotated objects across 1,268 RGB and IR image pairs. It includes nine different vehicle categories. For training and testing, we use images with a resolution of 1024×1024 . Evaluation on the VEDAI dataset is conducted using the mAP metric.

4.2 Implementation Details

Our method was implemented using PyTorch 1.8.1 on an Ubuntu 20.04 server, featuring a Xeon(R) Gold 5218R CPU, 128 GB RAM, and two Nvidia RTX 3090 GPUs, each with 24 GB of memory. The training process ran for 60 epochs with a batch size of 8. The initial learning rate was set to 0.01, utilizing the SGD optimizer with a momentum of 0.937. The weight decay is $5e-4$, with the learning rate adjusted via cosine annealing. Data augmentation techniques include mosaic and random flipping. The loss function is consistent with the original YOLOv5 framework, and the k parameter in the CAM is set to $2e-3$. In our ablation studies, YOLOv5 with NIN [11] serves as the baseline for comparison.

4.3 Quantitative Results

We evaluated the performance and generality of our MFCA module by performing experiments with the YOLOv5 detector using three different backbones: VGG16, ResNet50, and CSPDarkNet53. As shown in Table 1, our method achieves improvements of 1.08%, 1.58%, and 1.17% over the baseline for VGG16, ResNet50, and CSPDarkNet53, respectively on the KAIST dataset. On the FLIR dataset, it achieves improvements of 4.2%, 3.1%, and 4.0%. These findings demonstrate that the MFCA module is both versatile and effective across

different backbones and evaluation metrics. Additionally, it is clear that the detector utilizing CSPDarkNet53 outperforms the other methods, achieving the best overall results.

Table 1. Comparison with different backbones.

Backbone	Method	KAIST MR(%)	FLIR mAP50(%)
VGG16	Baseline	16.12	69.3
	Ours	15.04	73.5
ResNet50	Baseline	13.65	70.5
	Ours	12.07	73.6
CSPDarkNet53	Baseline	8.33	76.5
	Ours	7.16	80.5

KAIST Dataset. Table 2 presents a comparison of our method against other existing approaches on the KAIST dataset. The results show that, under a standard setup, our method exceeds most competitors’ performance, achieving the lowest miss rate in the combined case. Specifically, our method achieves 7.16%, 6.94%, and 6.38% in the MR_All, MR_Day and MR_Night metrics, respectively. In the integrated scenario, our method performs comparably to the state-of-the-art method ICAFusion. In the night scenario, MFCA is 1.47% lower than the ICAFusion. This indicates that MFCA provides significant advantages in low-light conditions, thanks to its ability to enrich semantic information.

Table 2. Comparison for different models on the KAIST dataset.

Model	Backbone	MR_All	MR_Day	MR_Night
ACF [6]	VGG16	47.32	42.57	56.17
IAF-RCNN [12]	VGG16	15.73	14.55	18.26
CIANet [29]	VGG16	14.12	14.78	11.13
Halfway Fusion [14]	VGG16	25.75	24.88	26.59
MLPD [9]	VGG16	7.58	7.96	6.95
MSDS-RCNN [11]	VGG16	8.23	8.83	6.75
CCIFNet [24]	Resnet50	7.21	7.65	6.84
MBNet [31]	Resnet50	8.40	8.62	8.27
ICAFusion [20]	CSPDarknet53	7.17	6.82	7.85
MFCA(Ours)	CSPDarknet53	7.16	6.94	6.38

FLIR Dataset. Table 3 presents a comparison on the FLIR dataset between our method and existing methods. The results indicate that our method surpasses the performance of most current approaches. Specifically, our approach achieves 80.5% in mAP50, 38.9% in mAP75, and 41.5% in the overall mAP metric. These results demonstrate that MFCA achieves competitive performance.

Table 3. Comparison for different models on the FLIR dataset.

Model	Backbone	mAP50	mAP75	mAP
CFR [26]	VGG16	72.4	—	—
GAFF [27]	VGG16	72.7	30.9	37.3
GAFF [27]	Resnet18	72.9	32.9	37.5
BU-ATT [8]	Resnet50	73.1	—	—
BU-LTT [8]	Resnet50	73.2	—	—
CFT [15]	CSPDarknet53	73.0	32	37.4
CFT [15]	CFB	78.7	35.5	40.2
ICAFusion [20]	CSPDarknet53	79.2	36.9	41.4
MFCA(Ours)	CSPDarknet53	80.5	38.9	41.5

Table 4. Comparison for different models on the VEDAI dataset.

Model	mAP50	mAP75	mAP
SuperYOLO [28]	75.1	—	—
ICAFusion [20]	76.6	36.9	44.9
YOLOFusion [16]	78.6	—	49.1
CFT [15]	79.7	—	46.8
MFCA(Ours)	80.2	40.7	49.8

VEDAI Dataset. Table 4 presents a comparison on the VEDAI dataset between our method and existing methods. Listing methods in the table utilise CSPDarknet53 as the backbone. MFCA achieves 80.2%, 40.7%, and 49.8% in the mAP50, mAP75, and mAP metrics, respectively. It demonstrates competitive performance compared to other state-of-the-art methods.

4.4 Qualitative Results

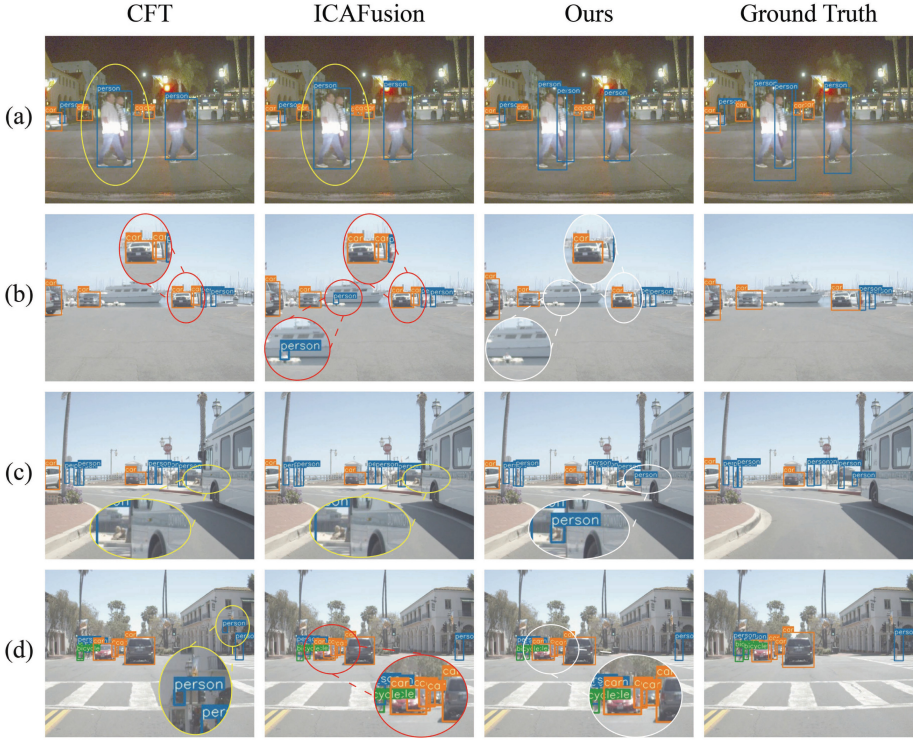


Fig. 5. A qualitative analysis comparing our method to others. The yellow ellipses represent instances of missing detection, while the red ellipses represent instances of erroneous detection. In order to facilitate the visualisation of the details, the image has been partially zoomed in. (Color figure online)

We perform a qualitative evaluation on the FLIR dataset in addition to the quantitative evaluation. Figure 5 illustrates the detection results for MFCA and related models. Since MFCA operates on the channel dimension, it enriches the semantic information. In the first set of results, two overlapping pedestrians are accurately detected separately. In the second set of results, the car with the door opened is identified as a single object, whereas in the comparison method, it is identified as two distinct objects. In the third set of results, the feature-rich RGB image features enable the accurate detection of the pedestrian in front of the bus. Similarly, in the fourth set of results, the complex boundary area is also accurately identified, with a low false alarm rate.

4.5 Ablation Study

This section provides ablation studies to assess the design of the MFCA model, which includes two sub-modules: the Spatial Calibration Module (SCM) and the Channel Aggregation Module (CAM). Each sub-module is assessed individually as a fusion component, with performance results detailed in Table 5. The results demonstrate that both SCM and CAM significantly improve prediction accuracy relative to the baseline. The SCM filters out redundant information from other modalities while incorporating valid feature information from other modalities as complementary elements. In contrast, the CAM addresses the inherent limitation of individual modalities from a channel perspective by compensating for the lack of semantic information.

Table 5. Comparison of mAP (%) when SCM or CAM is applied on the FLIR dataset.

SCM	CAM	mAP50	mAP75	mAP
		76.5	34.7	38.5
✓		78.6	35.6	40.5
	✓	77.9	35.4	39.8
✓	✓	80.5	38.9	41.5

5 Conclusion

We propose a method based on feature calibration and aggregation to improve the precision of multimodal object detection. Our study demonstrates that MFCA is an efficacious approach for acquiring information from both modalities, resulting in the calibration and aggregation of features across different modalities. Furthermore, it is demonstrated that both SCM and CAM can significantly enhance the detection precision. By integrating both location and semantic information across the spatial and channel dimensions, their combination can further optimise the detection effectiveness, leading to competitive detection outcomes.

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